INCH-POUND
MIL-DTL-24643/42C
22 August 2002
SUPERSEDING
MIL-C-24643/42B
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG size 20, 7 strands, and AWG Size No. 18, class B stranding, ASTM B8, (7 by 0.0152 inch). For LS1SU-36, 32 AWG Size 20 conductors and 4 AWG Size No. 18 conductors. For LS1SU-60, 60 AWG Size 20 conductors.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.017 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG No. 34 or No. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape. The four size AWG No. 18 conductors shall be numbers 1 through 4.

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Construction (non-watertight) (continued)

Sixth - The required number of single (see table I) cabled with a lay not greater than 15 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm well rounded cable.
Seventh - Binder tape, applied helically with overlap.
Eighth - Cross-linked polyolefin jacket. (See table I for thickness).

TABLE I. Details.

Military part no. M24643/42	Size	Conductor size AWG No.	Diameter over completed conductor (nominal) (inch)	Cable jacket thickness min avg (inch)	Overall diameter (inches) TYPE LS1SU	
					min	max
-01UN	36	18 20-7	0.127 0.115	0.075	0.910	0.985
-02UN	60	20-7	0.115	0.100	1.210	1.310

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - ohms/1000 feet at 25°C, maximum
ASTM B 286, size 20-7 ..... 10.93
Size AWG no. 18 (class B) 7.05
Voltage withstand - volts, root mean square, minimum
Conductor to shield 2200
Shield to shield 500
Insulation resistance - Megohms -1000 feet minimum
Conductor to shield 500
Component shield to shield 100
Conductor and shield continuity ..... No failure

Group A:
Visual and dimensional ..... No failure
Capacitance - at 1 kilohertz (kHz), pF/ft, maximum 72
Group B:
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C) 50
Cold bending, cable - mandrel diameter 12 times cable diameter ..... No damage
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250

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Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals (aged)
Insulation
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage ..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5 °C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45

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Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-040)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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